Bridging Graph Polynomials and Symmetric Tensors

TENORS Workshop 2

Lael John

KU Leuven

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Symmetric Tensor Recap

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- For vector spaces $V_1, V_2 \dots V_d$ over a field k, their tensor product is the vector space $V := V_1 \otimes \cdots \otimes V_d$, describing all the multilinear mappings from $V_1^* \times \cdots \times V_d^* \to k$. [7]

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- $T \in V$ is a symmetric tensor $\iff \sigma(T) = T \; \forall \, \sigma \in S_d$, where S_d is the permutation group of d elements. [3]
- For elementary tensors this looks like $\sigma \cdot (v_1 \otimes \cdots \otimes v_d) = v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(d)} = v_1 \otimes \cdots \otimes v_d$

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- Notation: $V^{\otimes d}$ denotes the tensor product of a vector space V with itself d times. $S^d(V) \subset V^{\otimes d}$ denotes the subspace containing all the symmetric tensors.
- Elementary symmetric tensors are of the form $v^{\otimes d}$.

Relationship with Homogenous Polynomials

Consider $k[x_1 \dots x_n]_d$, the set of all homogenous polynomials of degree d over k (implicitly included in this is the 0 polynomial). Then an established result [3] is that

$$k[x_1 \dots x_n]_d \cong S^d(k^n)$$

as k-vector spaces, by linearly extending the mapping of monomials

$$\mathbf{x}^\alpha := \prod_{i=1}^{n} x_i^{a_i} \mapsto \frac{1}{d!} \sum_{\sigma \in S_d} \sigma(\bigotimes_{i=1}^{n} e_i^{\otimes a_i})$$

Here α denotes the tuple $(a_1, a_2, \dots, a_n)$ where $a_i \geq 0 \; \forall \; i = 1, 2, \dots, n$, and for this mapping $|\alpha| = \sum_{i=1}^{n} a_i = d$.

Veronese Variety

Geometrically, consider $\mathbb{P}V$ to be the set of all lines in $V \cong k^n$ passing through the origin. Then the Veronese mapping given by

$$\nu_d : \mathbb{P}V \to \mathbb{P}k^N$$

taking points with homogenous coordinates

$$[x_1 : x_2 : \ldots : x_n] \mapsto [x_1^d : x_1^{d-1}x_2 : \ldots : x_n^d]$$

allows for the visualisation of elementary symmetric tensors as points on a variety in $\mathbb{P}k^N$. Here $N = \binom{n-1+d}{d}$ [10]. More general symmetric tensors then lie on secant varieties to the Veronese.

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- A *tree* is a connected graph with no loops i.e. $b_0(G) = 1, b_1(G) = 0$.
- A *spanning* tree $T \leq G$ contains all of V but not (necessarily) all of E.

Constructing (particular) Graph Polynomials

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For the rest of this presentation $G = (V, E)$ such that $\|G\| = n$ and $b_0(G) = 1$.A *Kirchoff* polynomial [6, 11] for a given graph G is given by

$$P_G(x_1, \ldots, x_n) := \sum_{T \leq G} \prod_{e \notin T} x_e$$

Note the variables of the polynomial correspond to a choice of labelling the edges.

Properties of the Kirchoff Polynomial

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- The degree of each Kirchoff polynomial is exactly $b_1(G)$
- Variables common to each term in the polynomial denote self-loops of G.
- An equivalence relation $\sim_S$ can be imposed by

$$\begin{aligned} P_A \sim_S P_B \quad &\iff \quad P_A(x_1, \dots, x_n) = \tau \cdot P_B(x_1, \dots, x_n) \\ &\iff \quad P_A(x_1, \dots, x_n) = P_B(x_{\tau(1)}, \dots, x_{\tau(n)}) \end{aligned}$$

Here $\tau \in S_n$ permutes n elements, corresponding to a relabelling of edges.

What is... the Grothendieck Ring of Varieties?

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- $\mathcal{K}_0(V_k)$ modulo all relations of the form $[X] = [X \setminus Y] + [Y]$ results in $\mathcal{K}(V_k)$, a free, additive abelian group.

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- This group can be seen as a commutative ring with unity by defining multiplication $[X][Y] = [X \times_k Y]$, the fibered product.
- $\mathcal{K}(V_k)$ can be seen as a module over $\mathbb{Z}[\mathbb{L}]$ where $\mathbb{L} := [\mathbb{A}^1]$ [1].

Belkale, Brosnan (2003)

Let $\mathbb{V}(P_G) \subset \mathbb{A}^n$ denote the variety given by zeroes of P_G. Define $Y_G = A^n \setminus \mathbb{V}(P_G)$.

Theorem (Belkale, Brosnan (2003) [2])

Take $S := \{\mathbb{L}^n - \mathbb{L}\}_{n \geq 2}$ as generating a multiplicative subset in $\mathcal{K}(V_k)$. Then the collection of $[Y_G]$ generate $S^{-1}\mathcal{K}(V_k)$ as a $S^{-1}\mathbb{Z}[\mathbb{L}]$ module.

Aluffi, Marcolli 2011

X, Y are stably birational $\iff$ $X \times \mathbb{P}k^m \dashrightarrow Y \times \mathbb{P}k^n$ is birational, for $n, m \geq 0$.

Theorem (Larsen, Lunts (2001) [8])

$\mathcal{K}(V_k)/(\mathbb{L}) \cong \mathbb{Z}[SB]$, the ring of stably birational varieties

Theorem (Aluffi, Marcolli (2011) [1])

The images of $[Y_G]$ in $\mathcal{K}(V_k)/(\mathbb{L})$ generate $\mathbb{Z} \subsetneq \mathbb{Z}[SB]$

Corollary

Suppose $S' := \{\mathbb{L}^n - 1\}_{n\geq 1}$ Then $[Y_G]$ do not generate $S'^{-1}\mathcal{K}(V_k)$ over $S'^{-1}\mathbb{Z}[\mathbb{L}]$

Graphs as a Disjoint Union

Let $\mathcal{G}_n$ denote the family of all connected graphs with a fixed number of edges n.

- The corresponding graph polynomials all lie in $R := k[x_1, \dots x_n]/(x_1^2, \dots, x_n^2)$.

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- $\mathbf{G}_n$ can then be expressed as the disjoint union $\bigsqcup_{i=0}^{n} \mathbf{G}_{n,i}$ where each component contains the graphs with $b_1(G) = i$.

Addition in Graphs?

In $\mathbf{G}_n$, an attempt to define an addition operation is given below

- Let $[A]_{KP,S}, [B]_{KP,S} \in \mathbf{G}_n := (\mathcal{G}_n/\sim_{KP})/\sim_S$

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- P_C can be chosen as an element with the largest number of summands in this set. $[P_C]_S$ then is a corresponding equivalence class.
- **Conjecture:** Does there always exist C, a graph, such that its Kirchoff polynomial is P_C? If yes then
- *This turns* $\mathbf{G}_n$ *into an additive abelian group*

Image of R in symmetric tensors

What does R (the ring of Kirchoff polynomials) look like under the isomorphism between homogenous polynomials and symmetric tensors?
(**Note:** k is no longer required to be $\mathbb{F}_2$)

- R describes a quotient ring of homogenous polynomials, it must also induce images in the space of symmetric tensors

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- Since none of the elements in R can be expressed as linear polynomials to the d-th power, they cannot be seen as points lying on the Veronese variety.
- Consider the mapping $\mu_d : \mathbb{P}V \to \mathbb{P}k^{\binom{n}{d}}$ taking $[x_1 : \ldots : x_n] \mapsto [\prod_{i=1}^{d} x_i : \ldots : \prod_{i=n-d+1}^{n} x_i]$.

Kirchoff Polynomials as Symmetric Tensors

$$
\begin{tikzcd}
\mathbb{P}V \arrow[rr, hook, "\nu_d"] \arrow[dr, hook, "\mu_d"'] & & \mathbb{P}k^{\binom{n+d-1}{d}} \\
& \mathbb{P}k^{\binom{n}{d}} \arrow[ur, hook, "i"'] &
\end{tikzcd}
$$

Thus Kirchoff polynomials can be seen as specific points on $\sigma_r(i(\mu_d(\mathbb{P}V))) \subseteq \sigma_r(\nu_d(\mathbb{P}V))$ where r denotes the rank of the image tensor under Φ.

Symmetric Tensors and the Grothendieck Ring

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- Every homogenous polynomial generates a variety (and its complement).
- Therefore every symmetric tensor can be associated to a variety (or its complement)
- In particular the symmetric tensors associated to graph polynomials then correspond to $[Y_G]$ as defined earlier as well as the complement of $\mathbb{V}(P_G)$.

Generation of Symmetric Tensors

What does the element $[\mathbb{A}^1] =: \mathbb{L} \in \mathcal{K}(V_k)$ *correspond to when working with symmetric tensors or homogenous polynomials?*

- $[\mathbb{A}^1]$ is the equivalence class of all varieties having a birational morphism to $\mathbb{A}^1$

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- In n variables, the intersection of $(n-1)$ of the $\mathbb{V}(x_i)$ is birational to $\mathbb{A}^1$
- Describing the intersection in terms of a single homogenous polynomials is infeasible.
- Open question: Does reframing this attempt in terms of prime ideals generated by these Kirchoff polynomials work?

Further Questions

- Is there more underlying structure to the quotient ring R of Kirchoff polynomials? E.g. for $n = 5$, the polynomial $x_1x_2 + x_3x_4 \in R$, but doesn't correspond to any graph G with $b_0(G) = 1$ and $b_1(G) = 2$

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- Does applying a grading on the space of graphs only work when dealing with the $\mathbb{F}_2$? Can this be extended to any field k?

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- Does applying a grading on the space of graphs only work when dealing with the $\mathbb{F}_2$? Can this be extended to any field k?
- If a BB type result could be extended to symmetric tensors - what would this indicate for questions of rank decompositions?

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